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Semi-Supervised Latent Variable Model for NYC Property Valuation

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Motivation & Problem

Property valuation underpins **tax administration** to ensure fair revenue and **urban planning** to optimize resource allocation. We target both: taxation requires precise financial estimates, while planning uses value gradients to identify development zones.

Challenges:

- **Heavy-tailed Distribution**: Prices exhibit excess kurtosis of 9.27, violating standard Gaussian assumptions.
- **Missingness**: About 25% of records lack price data.

Goal: Learn a latent representation to predict sale prices and provide uncertainty estimates without discarding incomplete data.

Methodology

We implement a **Semi-Supervised MIWAE** (SemiSupMIWAE) using a Student-t Mixture Prior.

Generative Model

- **Latent Space**: $z_i \sim \sum_{k=1}^{K} \pi_k \, t_\nu(\mu_k, \Sigma_k)$.
- **Covariates**: $x_i \mid z_i \sim \mathcal{N}\left(\mu_x(z_i), \operatorname{diag}(\sigma_x^2(z_i))\right)$.
- **Price Head**: $y_i \mid z_i \sim \mathcal{N}\left(\mu_y(z_i), \sigma_y^2(z_i)\right)$.

Inference We maximize the importance-weighted lower bound, the ELBO:

$$\mathcal{L} \approx \mathbb{E}_{q_\phi}\left[\log \frac{1}{K_{\text{IW}}} \sum_{k=1}^{K_{\text{IW}}} \frac{p_\theta(x_i, y_i \mid z_i^{(k)})\, p(z_i^{(k)})}{q_\phi(z_i^{(k)} \mid x_i, y_i)}\right]$$

Graphical Model

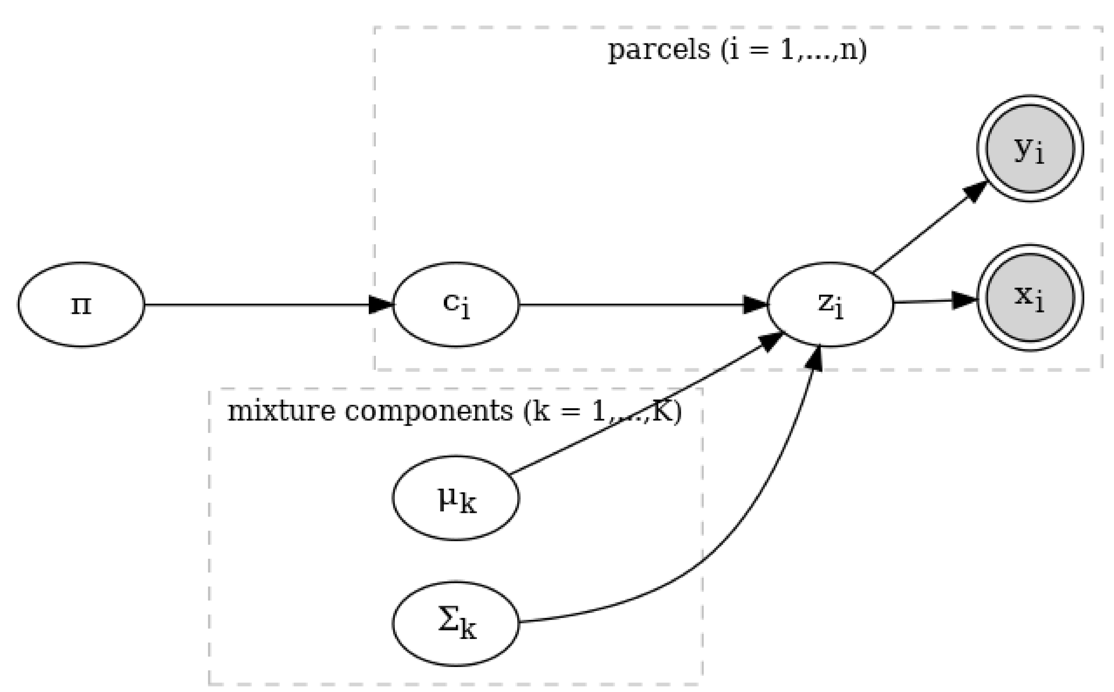

Figure: SemiSupMIWAE Structure.

Predictive Performance

Evaluated on a 20% held-out set.

Metric	Value
Avg. Log Posterior $p(y \mid x)$	-2.28
RMSE	$24.6 Million
MAE	$2.18 Million
Median Absolute Error	$128,000
95th Percentile Error	$60 Million

Table: Predictive metrics.

Residual Analysis

Spatial plotting reveals bias, while histogram confirms heavy tails (kurtosis = 9.27).

Latent Space Analysis

The 3D latent space, with $K = 3$, captures interpretable structure. We plot the top two dimensions by variance contribution: z_3 (49.9%) and z_1 (30.2%).

Price Gradient SHAP attribution reveals the primary latent dimension encodes recent market trends and cumulative sales volume.

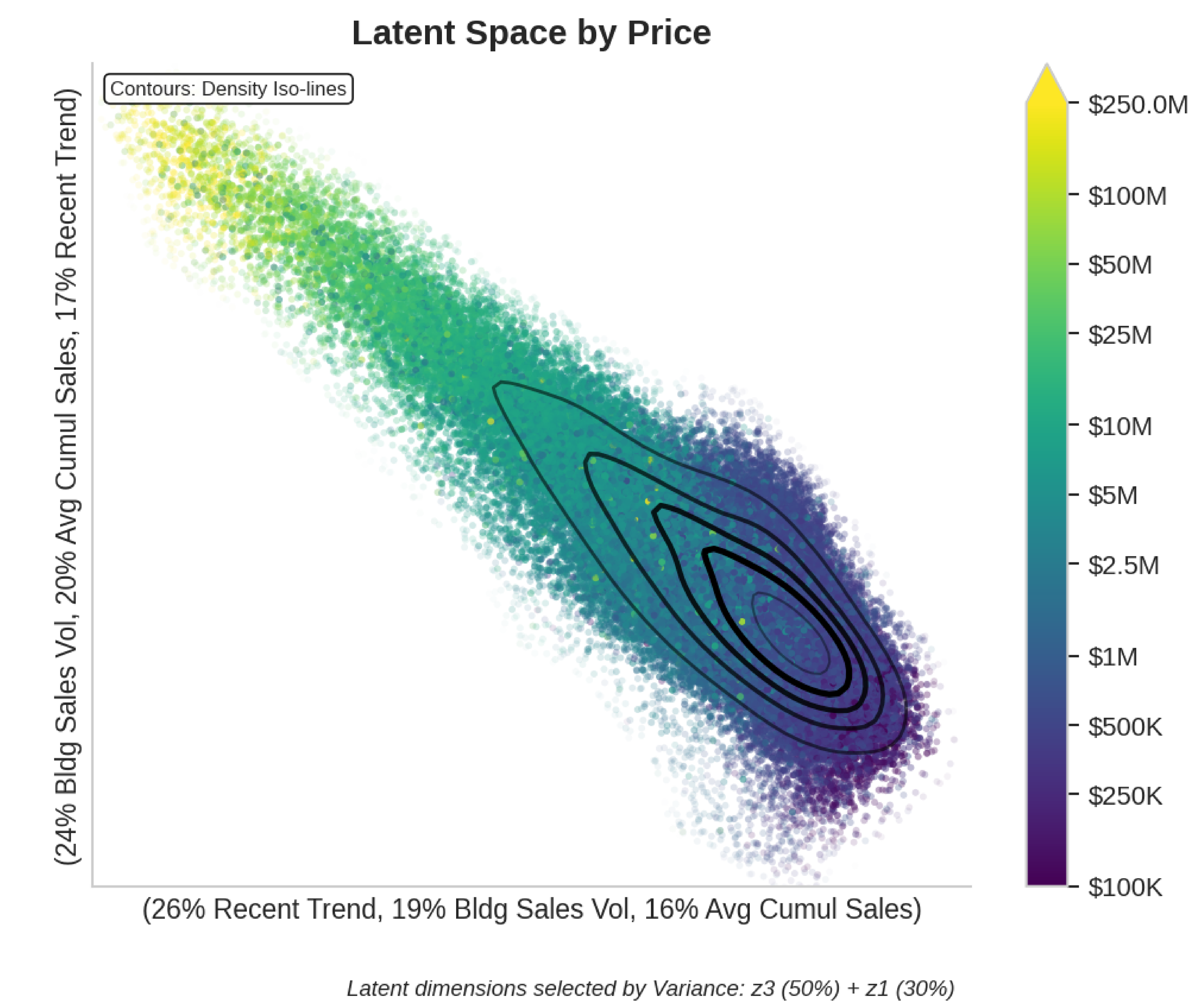

Figure: Latents colored by Price Decile.

Building Type Structure Building classes form identifiable striations rather than discrete clusters, reflecting the continuous nature of property characteristics.

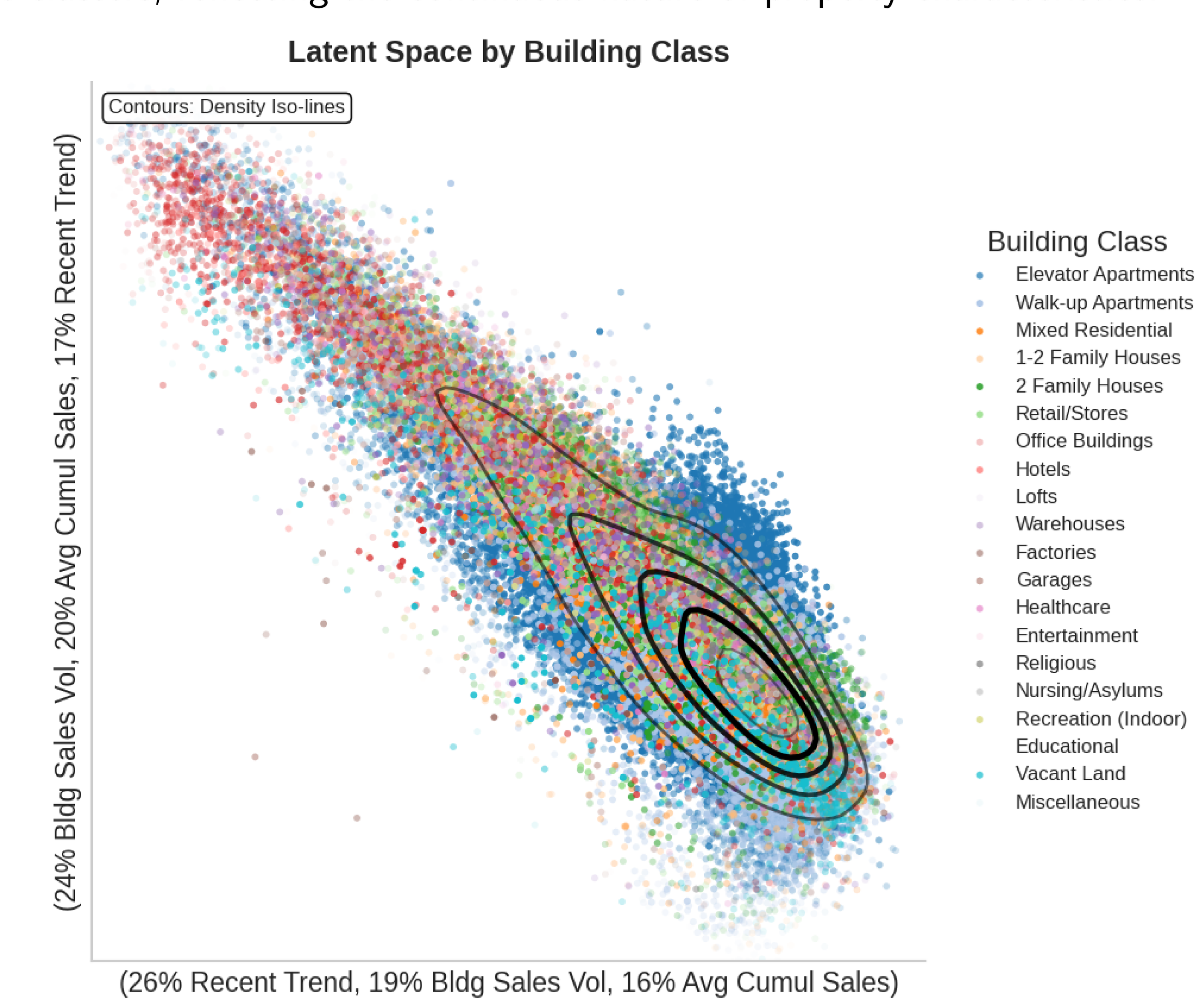

Figure: Latents colored by Building Class.